

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: An Imbalance-Dependent Boundary for Minority Retention

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Abstract

Self-consuming training loops can induce *model collapse*, including loss of rare modes. We ask when unlabeled likelihood ranking of synthetics harms minority modes in self-consuming DDPM training on imbalanced mixtures. The answer is **imbalance-dependent**: we report a moderate-imbalance boundary for the full bottom>rand>top ordering, not a universal law. Define $r = m^{(1)}/m^{(0)}$. At $\rho=5$ ($n=5$), generation 1 retention obeys bottom > rand > top on all seeds; bootstrap 95% CIs for mean paired Δ vs. random- k exclude zero (top-rand $[-0.37, -0.15]$; bottom-rand $[0.16, 0.43]$), while exact Wilcoxon $p=0.0625$ is the discrete floor at $n=5$ (directional). At intermediate $\rho=7$ the full order holds on only 2/5 seeds; at $\rho=10$ ($n=8$) on 3/8. Top-rand CIs still exclude zero at $\rho=7$ and $\rho=10$, but bottom-rand does not at $\rho=10$ —consistent with a partial floor on rankable minority mass. Strongest powered claim: top- k does *not* improve sliced W_2 at generation 1 (18/18 cells positive; bootstrap CI of mean Δ excludes zero; exact $p=7.6\times 10^{-6}$; Holm $p=3.8\times 10^{-5}$). Directional transfers (8D GMM, Digits-PCA) match $\rho=5$; Digits pixel-space DDPM and Digits-CVAE do not stably replicate the full order (honest negative/mixed image checks at toy resolution). Code: <https://github.com/muhammadalahamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. An *imbalance-dependent boundary*: likelihood-rank ordering of minority retention is clear at moderate imbalance and breaks down under severe imbalance—plus a falsification of “top- k looks better on W_2 ”—not a universal law or ImageNet-scale SOTA method.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Stratified evidence for a moderate-imbalance regime ($\rho=5$) with directional bottom>rand>top ordering, vs. null/floor behavior at severe imbalance. (3) Powered evidence that top- k does not improve sliced W_2 vs. random- k at generation 1. (4) Directional 8D GMM and Digits-PCA checks (Digits-CVAE inconclusive).

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Qiao et al. [2026] prove that selection against a *biased local* real-data reference can prune global tail modes and accelerate collapse in siloed settings; their remedy uses collaborative Wasserstein proxy references. We study a complementary failure mode: ranking synthetics by the *generator’s own* denoising-likelihood proxy, with a matched-budget top- k / random- k / bottom- k contrast and minority-mass retention as the primary diversity metric. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests use the **exact** Wilcoxon signed-rank distribution over sign assignments (enumeration; midranks for magnitude ties), not only a normal approximation.

4 Experiments

4.1 Primary design ($n=18$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`. We expand beyond the original eight cells by adding intermediate imbalance $\rho=7$ ($n=5$) and additional $\rho=10$ seeds ($n=8$ total) so the breakdown of the full ordering can be stated more sharply than on three seeds alone.

Table 1: Primary cells (seed $\times$ ρ).

$\rho \backslash$ seed	0	1	2	3	4	5	6	7
5	✓	✓	✓	✓	✓	—	—	—
7	✓	✓	✓	✓	✓	—	—	—
10	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Per- ρ mean r and seed-wise counts (generation 1). Full order = bottom>rand>top.

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	5	0.737	0.456	0.183	5/5	5/5 / 5/5
7	5	0.437	0.266	0.145	2/5	4/5 / 3/5
10	8	0.329	0.247	0.164	3/8	7/8 / 4/8

4.2 Retention: stratified primary analysis

Because strata differ, we do **not** treat a single pooled Wilcoxon as the primary claim. We lead with bootstrap 95% CIs of the mean paired Δ (more honest than p -values alone at these n); exact Wilcoxon p is reported alongside. Table 2 gives means and seed-wise order counts; Table 3 gives stratified paired contrasts.

At $\rho=5$, all five seeds agree in sign with large effects ($|d_z|>1.6$) and CIs excluding zero. Exact two-sided $p=0.0625$ is the smallest attainable at $n=5$ under the signed-rank null when all signs agree; the test *cannot* reject at $\alpha=0.05$. We report the $\rho=5$ *full* ordering as **directional** (unanimous signs + CIs), not as $p<0.05$. The full order is unreliable at $\rho=7$ (2/5) and $\rho=10$ (3/8). Top–rand CIs still exclude zero at $\rho=7$ and $\rho=10$ (top remains worse on average), but bottom–rand’s CI includes zero at $\rho=10$ —consistent with a partial floor on rankable minority mass under severe imbalance. Pooled $n=18$ tests are secondary (mix heterogeneous strata).

4.3 Sliced W_2

Generation-wise tests are repeated measures on the same cells (serially correlated), so Holm is a conservative acknowledgment rather than an assumption of independence. Evidence against top- k improving W_2 is strongest at $g=1$ (CI excludes zero; unanimous sign).

4.4 Real-mix α ablation (illustrative; $n=2$)

At $\alpha=0.5$, seed 0 breaks the full order (bottom $m^{(1)}$ slightly below random) while seed 1 holds—hence 1/2 (means still rank bottom>rand>top). At $\alpha=0.25$, likewise 1/2; at $\alpha=0.75$, top vs. random reverses on average (0/2).

4.5 Transfer checks (directional; $n=3$)

Minimum two-sided exact Wilcoxon p at $n=3$ with all signs agreeing is 0.25; we report Cohen’s d_z against that bound and do not claim powered generalization.

- **8D GMM:** 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3; top–rand $d_z = -2.56$, bottom–rand $d_z = +1.28$ (exact $p=0.25$).

Table 3: Stratified paired retention contrasts vs. random- k (bootstrap 95% CI of mean Δ ; exact Wilcoxon).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top-rand	-0.273	-1.96	$[-0.37, -0.15]$	0.0625
5	bottom-rand	+0.281	+1.65	$[0.16, 0.43]$	0.0625
7	top-rand	-0.121	-1.09	$[-0.21, -0.04]$	0.125
7	bottom-rand	+0.171	+0.92	$[0.03, 0.32]$	0.250
10	top-rand	-0.083	-0.79	$[-0.14, -0.01]$	0.078
10	bottom-rand	+0.082	+0.44	$[-0.03, 0.21]$	0.578

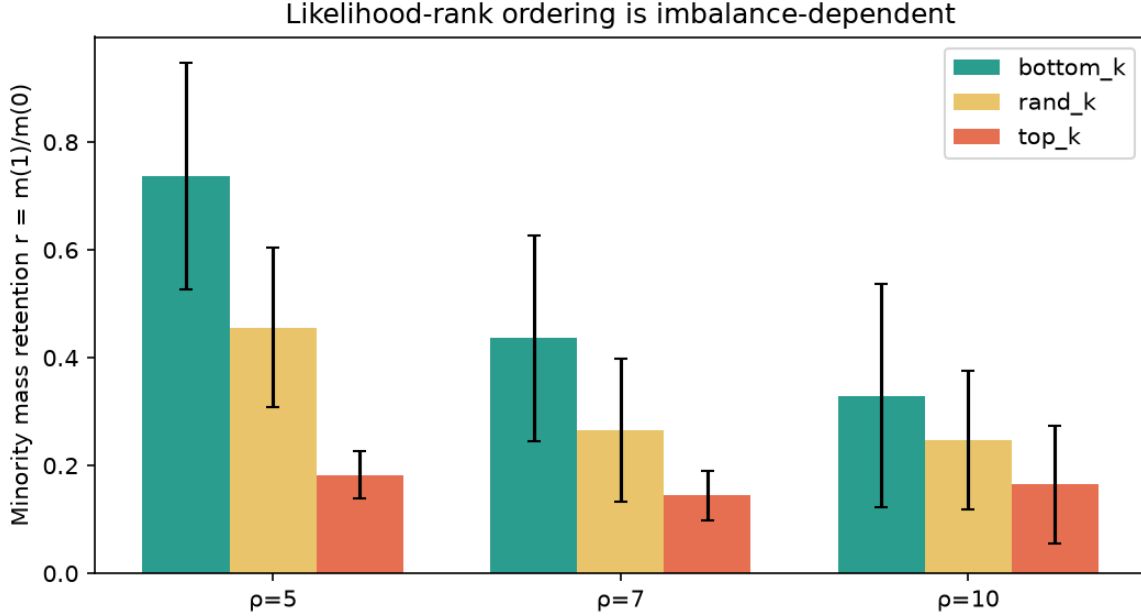


Figure 1: Primary retention r stratified by ρ (imbalance-dependent boundary).

- **Digits-PCA:** 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3; top-rand $d_z = -1.72$, bottom-rand $d_z = +2.67$ (exact $p=0.25$).
- **Digits pixel-space DDPM (8×8 MLP denoiser):** order 1/3 at $\rho=5$; means 1.25 / 0.88 / 1.23; bottom>rand on 3/3 seeds ($d_z = +1.27$) but top vs. random is inconsistent (exact signs 1/3 for top<rand). A small image-diffusion check, not ImageNet-scale validation; full ordering not replicated.
- **Digits-CVAE (inconclusive):** pixel-space conditional VAE did not reproduce a stable bottom>rand>top order.

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means; at 0.75, bottom vs. random flips (0/2 full order).

Table 4: $\Delta W_2(\text{top} - \text{rand})$ by generation on all 18 cells (positive = top worse). Bootstrap 95% CI; exact Wilcoxon p ; Holm across five generation-wise tests.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	Exact p (Holm)
1	+0.075	0.075	1.00	[0.044, 0.112]	18/18	7.6×10^{-6} (3.8×10^{-5})
2	+0.038	0.047	0.82	[0.019, 0.061]	16/18	0.0007 (0.0027)
3	+0.028	0.031	0.90	[0.014, 0.042]	16/18	0.0013 (0.0039)
4	+0.015	0.033	0.47	[0.000, 0.030]	14/18	0.010 (0.010)
5	+0.016	0.025	0.65	[0.007, 0.029]	15/18	0.0019 (0.0039)

Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$, default $k_{\text{frac}}=0.5$). “Order holds” = bottom>rand>top on $g=1$ minority mass. The $\alpha=0.5$ row is the *same cell* as $k_{\text{frac}}=0.5$ in Table 6.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	1/2
0.75	1.34	0.77	0.94	0/2

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics, reducing minority inflow. The *full* bottom>rand>top ordering is a moderate-imbalance phenomenon ($\rho=5$: 5/5), already unreliable at $\rho=7$ (2/5) and $\rho=10$ (3/8). Top remains worse than random on average even at higher ρ (top–rand CIs exclude zero), but bottom’s advantage fades as minority mass floors. Sliced W_2 is a poor monitor of minority survival: aggregate transport can look mixed while rare-mode retention collapses under top- k .

6 Limitations

Toy GMMs and small Digits image checks; not ImageNet-scale. At $\rho=5$, exact Wilcoxon cannot beat $p=0.0625$ with $n=5$ —retention inference leans on bootstrap CIs and unanimous direction; the powered negative result is W_2 at $g=1$ ($n=18$). α and k_{frac} ablations use two seeds; their default rows are the same experimental cell. Transfers: $n=3$ cannot beat exact $p=0.25$; Digits-CVAE was inconclusive. **Multiplicity:** Holm is applied within the W_2 generation family. Additional stratified/exploratory tests are not multiplicity-corrected paper-wide; we treat them as descriptive under the small-sample framing and lead with CIs. Contribution is a controlled empirical contrast complementary to MAD/LSF/Feng and to siloed-selection theory [Qiao et al., 2026]; venue target is workshop / short empirical note scope.

Exploratory search and interpretation of p -values. The reported claim is the survivor of an earlier exploratory search over several small experimental stacks (archived under `docs/` and `experiments/` in the public repository), not a single pre-registered hypothesis tested once. Readers should interpret retention p -values as descriptive under that history; we mitigate overclaiming by stratifying on ρ , demoting pooled tests, labeling $\rho=5$ full-order retention as directional, leading with bootstrap CIs, and treating the W_2 generation-1 result as the primary powered claim.

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$, $\alpha=0.5$). $0.25 \equiv 0.5$ when n_{syn} binds. The $k_{\text{frac}}=0.5$ row duplicates Table 5’s $\alpha=0.5$ row.

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

Likelihood-ranked synthetic selection induces an **imbalance-dependent** minority-retention pattern: a directional bottom>rand>top order at moderate imbalance ($\rho=5$), already unstable at $\rho=7$, and not seed-wise reliable at $\rho=10$. Top- k does not improve sliced W_2 vs. random- k at generation 1. Preferring “most likely” synthetics is not a free lunch for minority survival when rare modes remain measurable—and the effect should not be stated as a universal law.

References

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A Reproducibility

python scripts/reproduce_w2.py; stats tables via experiments/scripts/make_submission_stats.py.
Code: <https://github.com/muhammadalahzeb/likelihood-rank-self-consuming-diffusion>.